

Prioritized manuscript review: carpentry and joinery application

Date: 29 September 2026

Scope and audit method

This review covers the accessible chapter sources: `applications.tex` and `text/recurrent_scenarios/bn_explain.tex`, plus the resolved carpentry run and its Bayesian-network outputs.

The manuscript master file, bibliography, preamble, appendices, and most manuscript figure assets were not present in the reviewed workspace. The review therefore does not certify a complete LaTeX build or bibliography. It does verify internal references in the accessible TeX sources, the availability of graphics at the paths used by `applications.tex`, and the main reported numbers against the saved carpentry run.

The numerical audit found agreement for the main values: 417 accidents, 42 factors, 4,961 evaluated parent sets, 18 full-sample edges, 5,328 candidate scenarios, 905 observed candidates, and 80 closed scenarios at $n \geq 5$. The run records 500 bootstrap resamples and 9 of 18 full-sample edges with selection frequency ≥ 0.60 .

Overall assessment

The chapter has a strong analytical architecture. It separates empirical recurrence, association-rule measures, direct conditional dependence, bootstrap stability, and BN-implied recurrence. The role ordering makes the exact BIC optimization claim transparent, and the repeated non-causal qualification is appropriate.

The most likely reviewer concerns are reproducibility of the compiled manuscript, the operational rule for sparse CPTs, the completeness of stability reporting, transparency of selection choices, and the evidence offered for sensitivity to the recurrence threshold.

P0 - correct before circulation or submission

P0.1 Undefined method cross-reference

Evidence: `applications.tex`, lines 513-514, refers to `sec:chap4_global_bn_learning`. No such label occurs in the reviewed source set. The exact-learning subsection in `bn_explain.tex` is labelled `sec:chap4_global_bn_exact` at line 543.

Risk: the compiled PDF will contain an unresolved reference unless a separate unavailable source defines the label.

Action: replace the reference with `sec:chap4_global_bn_exact`, or add the intended label to the exact-learning subsection. Check the final PDF for all unresolved references.

P0.2 Missing graphic assets in the current compilation tree

Evidence: six of seven `includegraphics` paths in `applications.tex` are absent from `figs_ch4` in this workspace: `stability_landscape_all_roles.png`, `pareto_normalized_tchebycheff_all_roles.png`, `factor_resampling_A0.png`,

factor_resampling_A1_B_C.png, umap_seed_sensitivity_all_roles.png, and retained_factors_A1.png. The observed-versus-BN-implied figure is present.

Risk: this blocks a reproducible local build if the workspace reflects the submission tree.

Action: place the assets in figs_ch4, use a documented graphicspath, or add a build step that copies figures from the analysis run. Compile from a clean clone before submission.

P1 - major scientific reporting issues

P1.1 The CPT regularization decision rule is underspecified

Evidence: bn_explain.tex, lines 672-681, says MLEs are retained only if a "prespecified support criterion" is met. In applications.tex, lines 516-530, the smallest selected CPT count is 4 and two rows have count ≤ 5 , but MLEs are retained because no selected row is unobserved. The resolved configuration sets `cpt_regularization_min_observed_count` to 0.

Reviewer concern: the method sounds stricter than the rule used in the application. Sparse rows and boundary MLEs can affect model-implied scenario probabilities.

Action: state the operative rule exactly. Suggested wording: "Post-selection regularization was triggered only for an unobserved selected CPT row (threshold = 0). All selected rows were observed; MLEs were therefore used." Report the two rows with count ≤ 5 and the three boundary estimates as support limitations. Add a Jeffreys-prior sensitivity for the scenarios with the largest BN discrepancies, or justify why it is unnecessary.

P1.2 Stability is described, but its empirical result is not reported

Evidence: the subsection Bayesian-network estimation and stability assessment gives the bootstrap procedure and threshold, but no number of resamples or summary of the stable-edge result. The saved run used 500 resamples and has 9 of 18 full-sample edges with bootstrap frequency ≥ 0.60 .

Reviewer concern: the section title promises an assessment, yet the reader cannot judge the amount of stable structure without reopening the output.

Action: add this result sentence after the stability-threshold paragraph: "Across 500 accident-level bootstrap resamples, 9 of the 18 dependencies selected in the full sample had selection frequency at least 0.60." Include the generated stable-network figure and state that edge width encodes selection frequency and line style encodes the conditional contrast pattern.

P1.3 Threshold sensitivity does not yet test stability of principal scenarios

Evidence: bn_explain.tex, lines 93-97, says sensitivity is examined to determine whether principal recurrent configurations depend strongly on `n_min`. The application table reports only how many patterns remain at `n` $\geq 3, 5, 8$, and 10.

Reviewer concern: changing the number of retained scenarios does not show whether the substantive scenarios used for interpretation persist.

Action: retain the count table and add a nested-membership table for the highlighted scenarios, or a compact overlap analysis. State whether the

prevention and BN-discrepancy examples persist at $n \geq 8$ and $n \geq 10$.

P1.4 The eight labelled discrepancy scenarios need a stated selection rule

Evidence: the observed-versus-BN-implied figure and table identify S1-S8, but the current caption only says that labels refer to the table.

Reviewer concern: without the rule, the labelled examples can look selected after inspection.

Action: add a table note: "S1-S3 are the three largest positive count discrepancies; S4-S6 are the three largest negative discrepancies; S7-S8 are the nearest-to-equality scenarios among those with recurrence at least equal to the median." This is the rule used to generate the figure.

P1.5 Study-population selection needs an auditable definition

Evidence: applications.tex, lines 13-22, calls carpentry and joinery a sufficiently represented and occupationally specific group, but does not state the company activity codes, inclusion and exclusion rules, or a minimum-size criterion.

Reviewer concern: the family is analytically sensible, but selection can otherwise look discretionary and cannot be reproduced from EPICEA.

Action: provide the company-code family mapping in methods or appendix, state the eligibility rule, and give the number of accidents removed by each restriction. If the family was selected substantively rather than through a numerical threshold, say so plainly.

P2 - important interpretation and presentation improvements

P2.1 Show the stable network before scenario-level complementarity

Evidence: the stability section has a CPT-support table but no network result. A stable-network figure already exists in the saved run.

Action: place it at the end of the BN subsection. Use semantic labels, or retain factor IDs with a compact nearby key. The generated figure uses IDs only, which makes it hard to read without returning to the long inventory.

P2.2 Keep discrepancy descriptive, and add optional uncertainty support

Evidence: the text appropriately states that the discrepancy is not a p-value. However, 74 of 80 retained scenarios lie above the equality line.

Action: keep the current non-test language. In an appendix, add a bootstrap or held-out diagnostic for the distribution of observed-minus-implied counts, if feasible. Report it as robustness, not significance. Otherwise, avoid language suggesting a general lack of fit beyond this constrained, within-sample diagnostic.

P2.3 Make qualitative semantic review reproducible

Evidence: the factor section says representative factual units were reviewed qualitatively, but does not state how many units per factor were reviewed, how they were selected, or whether review was blinded to robustness metrics.

Action: add a short appendix protocol: number of representatives, ranking

rule, reviewers, and treatment of ambiguity. The existing cautious discussion of broad and mixed factors is a good basis for this addition.

P3 - clarity and economy

P3.1 Reduce numerical duplication around CPT support

The prose immediately before the CPT-support table repeats most table values. Keep the interpretive conclusion in prose and let the table carry the full counts.

P3.2 Standardize terminology after definition

The application alternates among configuration, pattern, and scenario. Define the technical distinction once, then use scenario for retained, interpreted configurations; reserve candidate configuration and closed pattern for their technical stages.

P3.3 Keep the revised expected-count language

The horizontal axis now uses "BN-implied expected recurrence, μ_s (accidents)", which correctly describes μ_s as an expected count. Maintain that wording in captions, tables, and prose.

Recommended revision order

1. Restore the missing reference and graphic assets, then run a clean complete manuscript build.
2. Specify the CPT regularization rule and report the 500-resample, 9-of-18 stability result with the stable-network figure.
3. Strengthen threshold sensitivity and state the figure-label selection rule.
4. Document company-code family selection and the semantic-review protocol.
5. Add optional discrepancy uncertainty diagnostics and make terminology edits.

Strengths to preserve

- The role-constrained ordering and proof of exact BIC optimization are clear.
- The distinction among recurrence, association, conditional dependence, bootstrap stability, and BN discrepancy is unusually careful.
- The chapter repeatedly rejects causal interpretation where the design cannot support it.
- The principal application counts checked in this review agree with the saved analysis outputs.